

ACIDS, BASES AND SALTS

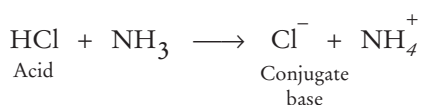
ACIDS

The substances which have sour taste and turn blue litmus to red are called **acids**.

- According to Arrhenius acids give hydrogen ion H^+ in an aqueous solution.



- According to Bronsted Lowry concept, acids donate proton (hydrogen ion).



(A conjugate acid-base pair differ by a proton)

- According to Lewis concept, acids have tendency to accept electrons, i.e. these behave as electrophiles. e.g. electron deficient species like BF_3 , $AlCl_3$, positive ions like Na^+ , K^+ and molecules having multiple bond between dissimilar atoms (e.g. CO_2 , SO_2 , etc).

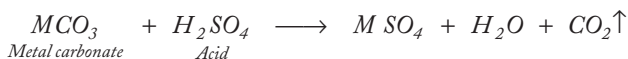
Sources of Some Acids

Acid	Sources
Citric acid	Lemon, orange, grapes (citrus fruits)
Maleic acid	Unripe apple
Tartaric acid	Tamarind, grapes
Lactic acid	Milk, curd
Acetic acid	Vinegar
Oxalic acid	Tomato, spinach
Hydrochloric acid	Stomach and other chemicals
Formic acid	Red ant sting

- Most of the acids contain hydrogen.
- HCl , H_2SO_4 , HNO_3 are mineral acids and are much stronger than organic acids.
- Aqua-regia is a mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio 3 : 1. It can dissolve inert materials like platinum, gold, etc.

Properties of Acids

- Acids liberate hydrogen gas with more reactive metals. e.g.
- $$\begin{array}{ccccccc} Zn & + & 2HCl & \longrightarrow & ZnCl_2 & + & H_2 \uparrow \\ \text{Metal} & & & & \text{Salt} & & \text{Hydrogen} \end{array}$$
- Acids evolve carbon dioxide (CO_2) gas with metal carbonates and bicarbonates. e.g.



- In aqueous solution, they are conductor of electricity due to the presence of free H^+ ions.

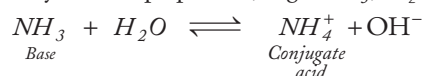
BASES

These substances which have bitter taste and turn red litmus to blue are called **bases**.

- According to Arrhenius, bases give hydroxyl ion (OH^-) in aqueous solution. e.g. $NaOH$, KOH , $CsOH$, $Mg(OH)_2$, etc.,



- According to Bronsted Lowry concept, bases have a tendency to accept proton, e.g. NH_3 , H_2O , etc.

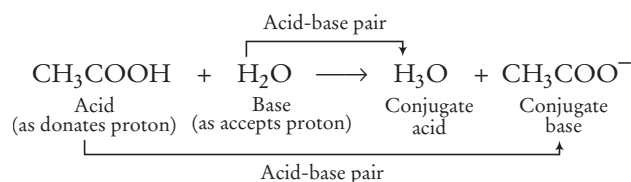


- According to Lewis concept, bases have a tendency to donate electron pair. e.g. simple anions like Cl^- , F^- , OH^- , etc., molecules containing lone pairs like $\ddot{N}H_3$, $R-\ddot{O}-H$, $R-\ddot{O}-R$, etc.

Properties of Bases

- Base reacts with metal to form salt and liberates hydrogen gas. However, such reactions are not possible with all metals.
- Water soluble bases are called **alkali**. Thus, all alkalis are bases but all the bases are not alkali.
- In aqueous solution and molten state, they are good conductors of electricity.

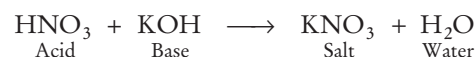
Acid and Base in a Reaction



- Paraldehyde (a carbonyl compound) is used as a hypnotic.
- Clove oil can be used as indicator.

SALTS

- When acid and base react together, they form salt and water. This reaction is known as **neutralisation reaction**, e.g.



- When acid and base both are strong, 13.7 kcal energy is released. However, if either the acid or the base is weak, energy released is less than 13.7 kcal because some part of energy is utilised to ionise weak acid or base.

- KNO_3 , K_2SO_4 , NaNO_3 , KCl , NaCl , KNO_3 are salts of strong acids and strong bases so, they cannot be hydrolysed and their aqueous solution is neutral.
- Solutions of salts of Cl^- , SO_4^{2-} , NO_3^- with metals other than Na^+ and K^+ are acidic. e.g. NH_4Cl . In other words, salt of a strong acid and a weak base undergoes hydrolysis in water to give acidic solution.
- Solutions of salts of Na^+ and K^+ with anions (except SO_4^{2-} , NO_3^- , Cl^-) are basic, e.g. CH_3COONa . In other words, salt of a weak acid and a strong base undergoes hydrolysis in water to give basic solution.

pH SCALE

SPL Sorensen suggested a scale to express the acidity or basicity of an aqueous solution in terms of hydrogen ion concentration which is known as pH scale. This scale is based on the ionic product of water.

The pH of solution is defined as “the negative logarithm to the base 10 of the hydrogen ion concentration in moles per litre”,
i.e. $\text{pH} = -\log [\text{H}^+]$

or in other words, the logarithm to the base 10 of inverse of H^+ ion concentration in the solution is its pH.
 $\text{pH} = \log \frac{1}{[\text{H}^+]}$

- The negative power to which 10 must be raised in order to express the H^+ ion concentration of a solution in moles per litre, is its pH.
i.e. $[\text{H}^+] = 10^{-\text{pH}}$
- In pure water or in neutral solution,
 $[\text{H}^+] = 10^{-7}$ so, $\text{pH} = 7$
- In an acidic solution, $[\text{H}^+] > 10^{-7}$ so, $\text{pH} < 7$
- In an alkaline solution, $[\text{H}^+] < 10^{-7}$ so, $\text{pH} > 7$

pH of Some Common Substances

Soft drinks	2.0-4.0
Tears	7.4
Lemon	2.2-2.4
Sea water	8.5
Human urine	4.8-8.4
Cow's milk, saliva	6.5
Rain water	6.0
Human blood	7.36-7.42

Importance of pH in Everyday Life

- Bee or ant sting (formic acid) leaves an acid into the skin which causes pain and irritation. Use of a mild base like baking soda (NaHCO_3) or calamine (zinc carbonate) is a remedy for it.
 - Wasp sting introduced some base, so a mild acid is a remedy.
 - Falling of pH of the mouth below 5.5 because of substances like chocolates and sweets lead to tooth decay which is naturally controlled by saliva (basic) to some extent but toothpaste being basic maintain this pH by neutralising excess acid.
 - Excessive acid (hydrochloric acid) secreted in our stomach which causes acidity, is neutralised by mild bases called antacids like milk of magnesia (or magnesium hydroxide) $[\text{Mg}(\text{OH})_2]$, aluminium hydroxide $[\text{Al}(\text{OH})_3]$, etc.
 - Every crop grows better in a particular pH range. e.g. slightly acidic soil is good for rice, neutral is good for sugarcane while citrus fruits generally grow in alkaline soil.
- ➔ **Note** Use of strong acid or base is not prescribed as these are corrosive in nature.
- Pickles are kept in glass jar because acid present in them reacts with the metal of metallic pot to give poisonous substances.
 - Acidity of soft drinks is regulated by bicarbonate salts.

pOH Scale

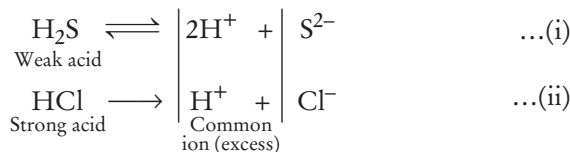
The pOH value of an aqueous solution may be defined as the negative logarithm of the hydroxide ion concentration,
i.e. $\text{pOH} = -\log [\text{OH}^-]$ and $\text{pH} + \text{pOH} = 14$.

➔ **Note**

1. When fresh ground water comes in contact of air its pH is slightly reduces. This is because of the presence of CO_2 in air which dissolves in water to give a weak acid, H_2CO_3 (carbonic acid). This acid provides H^+ ions and hence, increases the acidity of water due to which pH reduces.
2. pH of acidic solution increases when a base is added to it and pH of basic solution decreases when an acid is added to it.

Common Ion Effect

Addition of a strong electrolyte in the solution of weak acids or bases, decreases the ionisation of acid or base due to the presence of common ion. This is called common ion effect. e.g.



So, the reaction (i) goes in the reverse direction.

CHEMICAL THERMODYNAMICS AND SURFACE CHEMISTRY

THERMODYNAMICS

It deals with the study of heat and energy changes occurring during a physical and chemical processes.

An **open system** can exchange matter as well as energy with its surroundings, e.g. hot tea in a cup.

A **closed system** can exchange energy but not matter with its surroundings, hot water in a closed beaker.

An **isolated system** can exchange neither matter nor energy with its surroundings, e.g. thermos flask.

Thermodynamic Processes

Isothermal Process The process in which temperature of the system remains constant, i.e., $dT = 0$

Maximum work done during the isothermal expansion of 'n' moles of an ideal gas is given as

$$W_{\max} = -2.303 nRT \log \frac{V_2}{V_1}$$

Adiabatic Process The process in which no heat enters or leaves the system during any step of process, i.e. $dq = 0$.

Isobaric Process The process which takes place at constant pressure, i.e. $dp = 0$.

Isochoric Process The process in which volume of system remains constant, i.e. $dV = 0$.

Cyclic Process In a cyclic process, a system in a given state goes through a series of different processes but finally returns to its initial state, i.e. dE or $dU = 0$.

Spontaneous Process A process, i.e. physical or chemical change, which can proceed by itself is known as spontaneous process. There are also some processes which require some initiation before they proceed spontaneously, i.e. physical or chemical change.

- It is also known as feasible and probable process.
- The process is spontaneous, if free energy change, ΔG is negative.

SURFACE CHEMISTRY

Surface chemistry deals with the phenomenon that occurs at the surface or interfaces.

Catalysis

The term catalyst is given by Berzelius. A catalyst is a substance which alters the rate of reaction without being consumed in it and the phenomenon is known as **catalysis**.

General Characteristics of Catalysts

- A catalyst remains unchanged in mass and chemical composition during a reaction.

- Only a very small amount of catalyst is sufficient to catalyse a reaction. A catalyst does not initiate a reaction.
- A catalyst does not change the equilibrium state of a reversible reaction as it increases the rate of forward as well as backward reaction to the same extent.
- When a reversible reaction is performed in a closed container, a state is reached when the rate of forward and backward reactions become equal. This state is called **chemical equilibrium** or equilibrium state.
- Free energy change (ΔG°) at equilibrium is zero.

Types of Catalyst

1. **Positive and Negative Catalysts** A catalyst which increases the rate of a reaction (in forward direction) is called a **positive catalyst** while which decreases the rate of a reaction is called as **negative catalyst**.
2. **Induced Catalyst** The product of one reaction acts as catalyst for another reaction is called **induced catalyst**.
3. **Catalytic Promoters and Inhibitors** The substance which increases the activity of a catalyst are called **promoters** and which decreases the activity of a catalyst are called **inhibitors**.
4. **Catalytic Poisons** Catalytic poisons destroy the activity of a catalyst completely.

Enzyme Catalysis

The increase in the rate of reactions by the enzyme is known as enzyme catalysis. Since, they increase the rate of reaction occurring in living organisms, they are known by the name **biocatalysts**.

- All enzymes are protein in nature. Enzymes are highly specific in their nature. They are highly sensitive to temperature, i.e. the temperature at which the activity of enzymes is maximum, varies between 25-37°C.
- The rates of enzymatic reactions are very much affected by pH change.

Some Important Catalyst used in Different Processes

Processes	Catalyst used
Manufacture of ghee from vegetable oils	Nickel
Haber's process for the manufacture of ammonia	Fe (here Mo, acts as an activator)
Contact process for the manufacture of sulphuric acid	Pt powder
Conversion of proteins into peptide	Pepsin enzyme
Conversion of proteins into amino acids	Erepsin enzyme
Conversion of glucose into ethanol	Zymase enzyme
Conversion of starch into maltose	Diastase enzyme
Formation of vinegar from cane sugar	<i>Mycoderma aceti</i>
Conversion of sucrose into glucose and fructose	Invertase enzyme
Conversion of milk into curd	Lactase (lactobacilli)
Ostwald's process for the manufacture of nitric acid (HNO_3)	Platinum (Pt)

COLLOIDS

In a colloid, the size of solute particles is bigger than that of true solution but smaller than that of suspension, i.e. lies in between 1 to 100 nm. A colloidal solution is a heterogeneous system in which one substance is dispersed as very fine particles in another substance, called dispersion medium.

Properties of Colloidal Solutions

The important properties of the colloidal solutions are given below

Brownian Movement The rapid zig-zag movement executed by colloidal particle is termed as Brownian movement.

Tyndall Effect When a strong and converging beam of light is passed through a colloidal solution, its path becomes visible due to scattering of light by particles. It is called **Tyndall effect**. Blue appearance of sea water is an example of Tyndall effect.

Electrophoresis The phenomenon involving the migration of colloidal particles under the influence of electric field towards the oppositely charged electrode, is called electrophoresis.

➡ Colloidal solutions are purified by dialysis.

Emulsions

These are the colloidal solutions of two immiscible liquids in which the liquids act as the dispersed phase as well as the dispersion medium.

Types of Emulsions

There are two types of emulsions

- **Oil in water type** e.g. milk in which tiny droplets of liquid fat are dispersed in water.
- **Water in oil type** e.g. stiff greases, in which water being dispersed in lubricating oil.

Emulsifying Agents

During the preparation of emulsion, a small amount of some substances such as soap, gum, agar and protein etc are added to stabilise the emulsion. These substances are known as emulsifying agents.

- **Gel** is a colloidal system in which dispersed phase is liquid and dispersion medium is solid. e.g. butter, cheese. When a gel is allowed to stand for some time, it loses the dispersed phase (liquid), this phenomenon is called **syneresis** or **weeping of the gel**.
- Fog, clouds, mist are the colloidal systems of liquid in gas.
- Smoke is a colloidal systems of carbon (solid) in air (gas).
- Foams are the colloidal systems in which a gas is dispersed in liquid. e.g. alcohol, castor oil, shaving cream etc.
- Digestion of fats in intestine is aided by emulsification.
- **Coagulation** is the process of precipitation of colloid when it is shaken with some electrolyte. Its applications are
 1. Purification of muddy water by alum.
 2. Ferric chloride and alum are applied on wound to stop bleeding.

➡ **Note**

1. Blue colour of sky is due to the scattering of light by dust particles present in air.
2. The order of energy is 1 calorie > 1 joule > 1 erg.
3. Milk is a natural emulsion while paint is an artificial emulsion.

ELECTROCHEMISTRY

It is the branch of chemistry which deals with inter-conversion of chemical energies and electrical energies. It can be converted into electrical energy by means of an **electrochemical cell** (electrolysis).

Important Terms

- Chemical changes involving production or consumption of electricity are called **electrochemical changes**.
- The substance, which allow the electricity to pass through them are called **electrolytes**. e.g. common salt (NaCl), water (H₂O) etc.
- The substance which do not allow the electricity to pass through them are called **non-electrolytes**. e.g. sugar, wax, naphthalene etc.
- A number of metals such as Na, Mg, Ca and Al and a number of chemicals such as NaOH, Cl₂, Fe etc. are commercially produced by electrochemical methods.

Electrochemical Cell

It is a device, that produces an electric current from energy released by a spontaneous redox reaction (in short, which converts chemical energy into electrical energy).

This kind of cell includes the galvanic cell or voltaic cell. It has two conductive electrodes, i.e. anode or negative electrode (at which oxidation occurs) and cathode or positive electrode (at which reduction occurs). Electrolyte is filled in between the electrodes and contains freely moving ions.

BATTERY

An arrangement of one or more cells connected in a series, is called a battery. It is basically a galvanic cell.

Batteries are of two types :

1. **Primary Batteries** (non-rechargeable) These batteries act as galvanic cell and used only once. e.g. dry cell, mercury cell etc.

2. **Secondary Batteries** (rechargeable) These act as galvanic as well as electrolytic cell and can be reused. e.g. lead storage battery, nickel-cadmium battery.

Some Important Batteries

Battery	Anode	Cathode	Electrolyte	Used in
Leclanche cell (Dry cell)	Zinc	Carbon (Graphite) rod	Paste of ammonium chloride (NH_4Cl) and zinc chloride (ZnCl_2)	Transistors, clocks
Mercury cell	Zinc-mercury amalgam	Paste of HgO and C	Paste of KOH and ZnO	Hearing aids watches and camera
Lead storage cell	Lead	Lead packed with lead dioxide	38% solution of sulphuric acid	Automobiles, invertors
Nickel Cadmium cell	Cadmium	Nickel Dioxide	OH Solution	Torchlights, Electric Shavers

FUEL CELLS

These are galvanic cells, which use energy of combustion of fuels like hydrogen (H_2), methane (CH_4), methanol (CH_3OH) etc., as the source to produce electrical energy. e.g. hydrogen-oxygen fuel cell. This cell was used for the first time in **Apollo space programme**. The byproduct of this cell, i.e. water vapours, were condensed and used by the astronauts for drinking.

Advantages associated with fuel cell are

- Fuel cells are more efficient than conventional power stations or batteries etc.
- With a fuel cell there are fewer stages in producing the useful energy, so there is less opportunity to lose potentially useful energy, e.g. waste heat, friction from moving parts etc.
- A hydrogen-oxygen fuel cell is a non-polluting clean fuel since, the only combustion product is water.

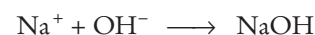
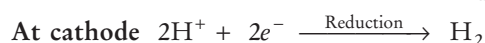
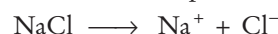
Disadvantages associated with fuel cell are

- Fuel cells cannot be used for large scale energy production, so conventional fossil fuels or nuclear power stations still have an important future.
- Hydrogen is a gas and requires a much larger storage volume as compared to fossil fuels like petrol.
- Safe storage is also an issue, especially as it would be stored under high pressure to decrease the storage space required. This immediately makes leaks and, accidents more likely to happen because hydrogen is a highly flammable explosive gas, too easily ignited.
- Efficient large scale technology is not yet developed to produce hydrogen on large scale, e.g. from electrolysis using solar power electricity, photovoltaic power system, wind turbines or hydroelectric power.
- Although water is cheap and plentiful, it requires expensive electrical energy to electrolyse water to split it into hydrogen and oxygen.

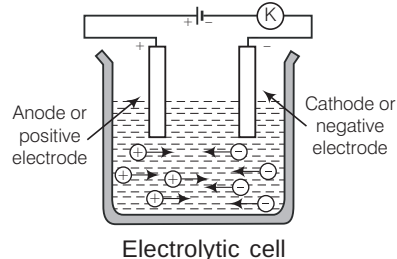
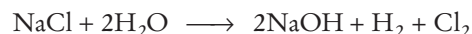
ELECTROLYSIS

The process, in which a non-spontaneous reaction is carried out by using electrical energy is called electrolysis and the instrument or container, in which the process of electrolysis is done, is called **electrolytic cell**.

Electrolysis of brine (the water saturated or nearly saturated with salt, usually sodium chloride) gives hydrogen and chlorine. The products are gaseous.



Net reaction



- **Note** During electrolysis, electrolytes get decomposed at the electrodes.

Electrolysis is used in the following ways :

- In the production of oxygen for spacecraft and nuclear submarines.
- In layering metals to fortify them.
- In production of hydrogen for fuel.
- In electrolytic etching of metal surfaces like tools or knives with a permanent mark or logo.
- In electroplating, electrorefining, electroprinting, electro-metallurgy and industrial preparation.

FARADAY'S LAWS OF ELECTROLYSIS

First Law of Electrolysis

It states that "The amount of electricity (or charge) required for oxidation or reduction depends upon stoichiometry of the electrode reaction."

$$w \propto Q, \quad w = ZQ = Zit$$

$$\therefore \text{Charge (Q)} = \text{Current (i)} \times \text{time (t)}$$

$$1 \text{ Faraday} = 96500 \text{ C mol}^{-1}$$